



Non-contrast CT-based radiomic signature for screening thoracic aortic dissections: a multicenter study

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Abstract

Objective To develop a non-contrast CT-based radiomic signature to effectively screen for thoracic aortic dissections (ADs).

Methods We retrospectively enrolled 378 patients who underwent non-contrast chest CT scans along with CT angiography or MRI from 4 medical centers. The training and validation sets were from 3 centers, while the external test set was from a 4th center. Radiomic features were extracted from non-contrast CT images. The radiomic signature was created on the basis of selected features by a logistic regression algorithm. The area under the curve (AUC) of the receiver operating characteristic (ROC) curve, accuracy, sensitivity, specificity, positive predictive value (PPV), and negative predictive value (NPV) were conducted to assess the predictive ability of radiomic signature.

Results The radiomic signature demonstrated AUCs of 0.91 (95% confidence interval [CI], 0.86–0.95) in the training set, 0.92 (95% CI, 0.86–0.98) in the validation set, and 0.90 (95% CI, 0.82–0.98) in the external test set. The predicted diagnosis was in good agreement with the probability of thoracic AD. In the external test group, the diagnostic accuracy, sensitivity, specificity, PPV, and NPV were 90.5%, 85.7%, 91.7%, 70.6%, and 96.5%, respectively.

Conclusions A radiomic signature based on non-contrast CT images can effectively predict thoracic ADs. This method may serve as a potential screening tool for thoracic ADs.

Key Points

- The non-contrast CT-based radiomic signature can effectively predict the thoracic aortic dissections.
- This radiomic signature shows better predictive performance compared to the current clinical model.
- This prediction method may be a potential tool for screening thoracic aortic dissections.

Keywords Radiomics · Aortic dissection · Tomography, X-ray computed · Screening

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Abbreviations

AD	Aortic dissections
AUC	Area under the curve
CI	Confidence interval
CT	Computed tomography
ESC	European Society of Cardiology
GLCM	Gray-level co-occurrence matrix
ICC	Interclass correlation coefficients
LASSO	Least absolute shrinkage and selection operator
MRI	Magnetic resonance imaging
mRMR	Max-Relevance and Min-Redundancy
NPV	Negative predictive value
PPV	Positive predictive value
Rad-score	Radiomics score
ROC	Receiver operating characteristic
ROI	Region of interest
TOE	Transesophageal echocardiography
TTE	Transthoracic echocardiography

Introduction

Aortic dissection (AD) is a rare but life-threatening condition associated with high morbidity and mortality rates [1]. The current incidence of acute AD was about 3 cases per 100,000 people, per year [2–4]. However, the true incidence of AD is unknown and as many as one-third of ADs may be undiagnosed [5]. The early diagnosis and treatment of ADs can reduce the downstream complications and prolong survival for those affected [6, 7]. The diagnosis of AD in the non-emergency setting would be of great benefit to patient outcomes. Among the existing imaging modalities, non-contrast CT is a widely used examination and may be a suitable imaging modality for screening ADs [1, 8–10]. Although non-contrast CT was found to be valuable for ruling in acute AD, the false-negative rate of non-contrast CT was high for ruling acute AD out because it had to be the minimum possible [11]. Therefore, it is valuable to create a non-contrast CT-based radiomic signature, which can be used to widely and objectively screen for thoracic AD.

Radiomics is the practice of high-throughput extraction of quantitative features to convert images into mineable data for guiding clinical decisions [12, 13]. Currently, the application of radiomics in cardiovascular imaging has lagged behind other fields, such as oncology [14]. Malignant tumors are commonly heterogeneous at both the genetic and histopathological levels, and the tumor heterogeneity can be quantified by radiomics on imaging [15, 16]. Several studies have also utilized radiomics to analyze blood heterogeneity to predict the enlargement potential of intracranial hematomas [17, 18]. The blood heterogeneity of aorta is affected by multiple elements in whole blood, the physiological parameters, and processes of blood flow complex parameters dictating blood flow, such as flow resistance, wall shear stress, pressure, wall

remodeling, and mass transport from the blood to the vessel wall [19–21]. Radiomics may offer a powerful large-data-based approach to quantitatively analyze the blood heterogeneity of the aorta on non-contrast CT images.

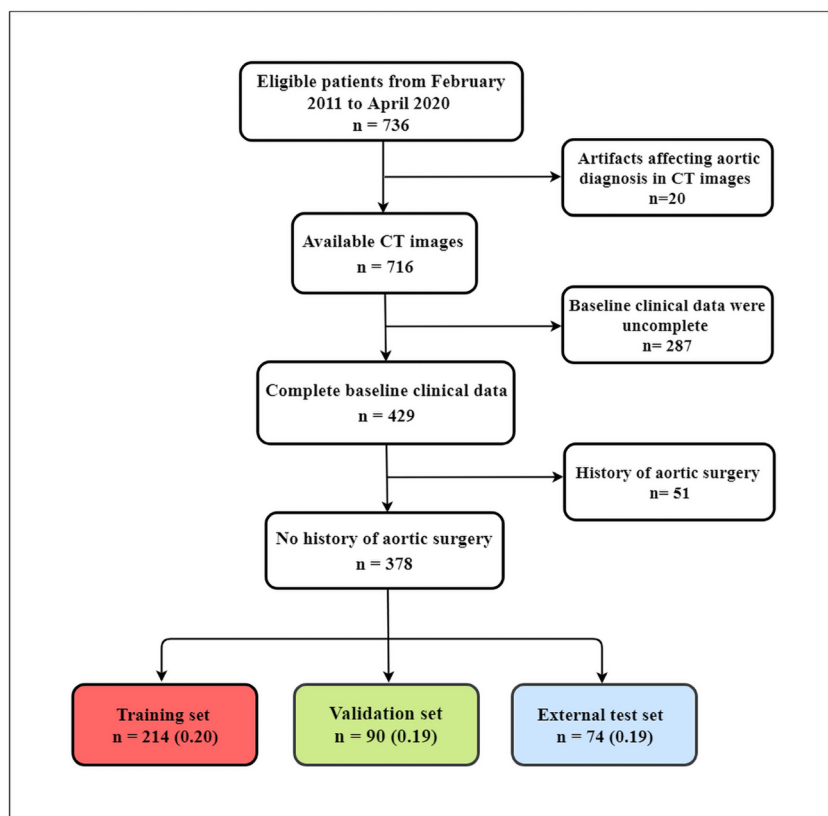
ADs also demonstrate an intricate and complicated flow pattern through the false and true lumens of the wall defect [20]. The blood heterogeneity of AD should be significant and quite different from that of the normal aorta and in other aortic conditions. In the present study, it was hypothesized that the non-contrast CT-based radiomics analysis of blood heterogeneity can distinguish thoracic AD from non-thoracic AD (normal thoracic aorta and other thoracic aortic conditions).

Patients and methods

Study participants

Our institutional ethics review board has approved this retrospective study, and waived the requirement for informed consent. A retrospective review of 736 patients who underwent aortic CT angiography or magnetic resonance imaging (MRI) in four hospitals from February 2011 to April 2020 was conducted (Fig. 1) including 100 patients from the First Affiliated Hospital of Zhejiang Chinese Medical University; 259 patients from Affiliated Jinhua Hospital, Zhejiang University School of Medicine; 255 patients from the Second Affiliated Hospital and Yuying Children's Hospital of Wenzhou Medical University; and 122 patients from the Wenzhou Third Clinical Institute Affiliated To Wenzhou Medical University. The inclusion criteria were as follows: (1) patients who underwent a non-contrast chest CT; (2) the diagnosis of each case was retrospectively confirmed by contrast-enhanced CT or MRI images; (3) the interval between non-contrast CT and contrast-enhanced CT or MRI was set to be less than 3 days according to the relevant follow-up recommendations in European Society of Cardiology (ESC) guidelines [9]. The exclusion criteria were as follows: (1) a history of aortic surgery; (2) artifacts affecting aortic diagnosis in CT images; (3) incomplete baseline clinical data including age, gender, hypertension status, smoking history, and chief complaints. Non-contrast CT images were evaluated by two experienced radiologists (reader A, with 5 years of experience in cardiovascular radiology; reader B, more than 20 years of experience), with independent attention to the following dichotomous variables: the imaging diagnosis of the aorta, displaced intimal flaps, displaced intimal calcifications, and high-attenuation areas. The radiologists were blinded to the clinical data and diagnosis when reviewing the images. In this study, the thoracic AD was defined as a positive class. The normal thoracic aorta and other thoracic aortic conditions were classified as non-thoracic AD and defined as the negative class. A total of 378 patients (median age, 63 years; interquartile range,

Fig. 1 Flowchart shows recruitment pathway for patients. Data in parentheses show the prevalence rate of thoracic AD



50–72 years; 114 females, 264 males, 139 emergency patients) were involved in this study. In addition, 358 patients were excluded from our study. Seventy-four patients from the

Wenzhou Third Clinical Institute Affiliated To Wenzhou Medical University were used as an external test set for the final assessment. According to the diagnosis of each case, 304

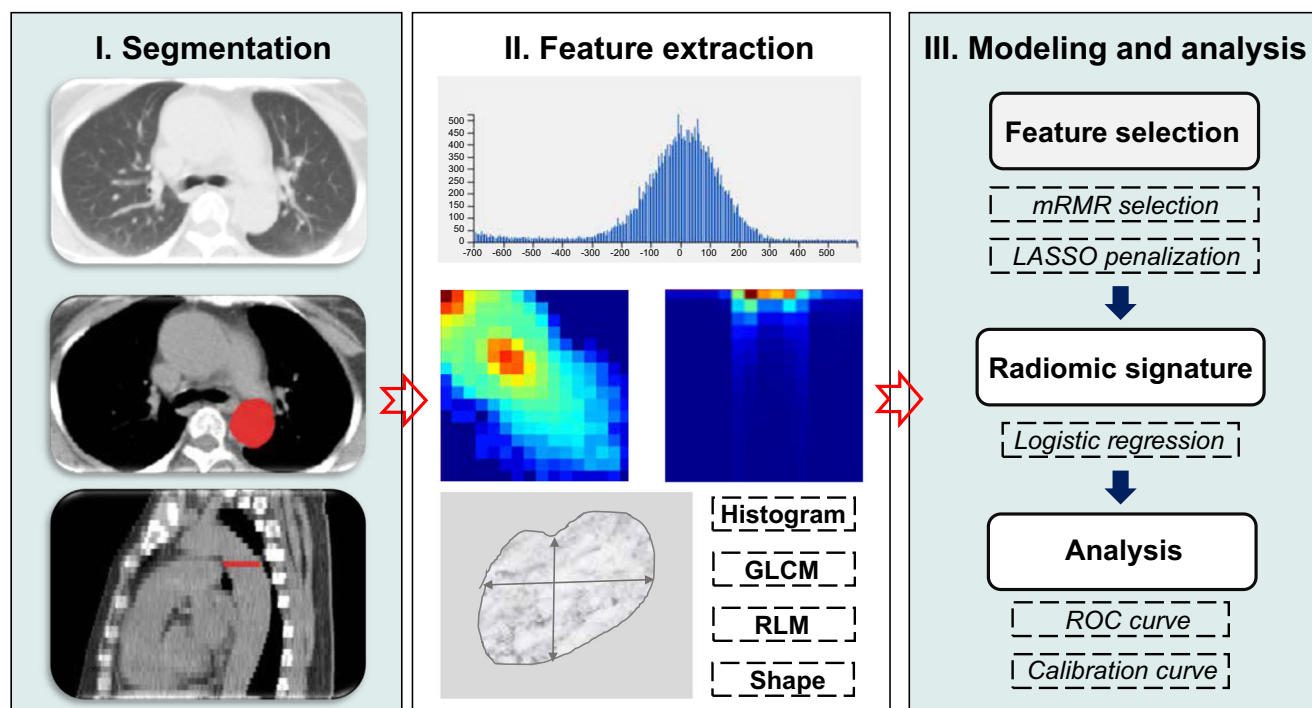


Fig. 2 Workflow of necessary steps for radiomic analysis. *mRMR* Max-Relevance and Min-Redundancy, *LASSO* least absolute shrinkage and selection operator, *GLCM* gray-level co-occurrence matrix, *RLM* run-length matrix, *ROC* receiver operating characteristic

patients from the remaining three centers were enrolled in the training and validation sets based on stratified sampling (214 and 90 patients, respectively). The flow diagram of our study is shown in Fig. 2.

Image acquisition

Non-contrast chest CT imaging was employed for radiomic analysis. Images of training and validation sets were acquired from three multi-slice CT scanners (Brilliance 16 CT or Brilliance iCT, Philips Healthcare; Definition AS, Siemens Healthcare). Images of the external test set were obtained from three multi-slice CT scanners (Optima CT680, GE Healthcare; Sensation 16 or Definition AS, Siemens Healthcare). The main scanning parameters of the Brilliance 16 CT scanner were as follows: tube voltage, 120kVp; tube current, automatic tube current modulation; matrix size, 512×512; reconstruction filter, filtered back-projection; slice thickness, 5 mm. The reconstruction filter “B70f” was used with the Siemens Somatom Definition AS CT scanner in both centers. The other imaging parameters such as the tube voltage, tube current, matrix, and slice thickness were consistent with Brilliance 16 CT.

Segmentation and radiomic feature extraction

The regions of interest (ROIs) of CT images were manually segmented on ITK-SNAP 3.6.0 [22]. We selected non-contrast CT images at the level of the carina for ROI segmentation of descending aorta. The carina plane was defined as the level where the trachea bifurcates to the left and right main stem bronchi [23]. Aortic calcification was not included in the ROI. A total of 396 radiomic features were extracted using AK software (Analysis Kit, GE Healthcare). The extracted features included histograms, texture features, form factor features, gray-level co-occurrence matrix (GLCM), gray-level run-length matrix (RLM), and gray-level size zone matrix (GLZSM). The extracted features are detailed in Appendix E1 (online).

Intra- and inter-observer agreement

The intraclass correlation coefficient (ICC) was used to evaluate the intra- and inter-observer agreement for feature extraction. Initially, ROI segmentation was performed by a radiologist (reader C, more than 5 years of experience in cardiovascular radiology). Thirty random CT images were chosen to evaluate the intra- and inter-observer agreement. The radiologist repeated the same procedure in 30 random CT images 2 weeks later to evaluate the intra-observer agreement. The thirty random CT images were manually segmented by another radiologist (reader D, more than 10 years of experience in cardiovascular radiology) to evaluate the inter-observer

agreement. A value of ICC is greater than 0.75 was considered in good agreement.

Statistical Analysis

All statistical analyses were carried out with R software, version 3.5.2 [24]. Differences in clinical factors and CT manifestations between thoracic AD patients and non-thoracic AD patients were assessed using an independent *t*-test or Mann-Whitney *U* test for continuous variables, and a Fisher’s exact test or chi-square test for categorical variables. A significant result was considered when the *p* value was less than 0.05.

The extracted features with an ICC of > 0.75 were considered possible parameters for the radiomic signature construction. In this study, ICC estimates were calculated based on a single-measurement, absolute-agreement, 2-way random-effects model. Max-Relevance and Min-Redundancy (mRMR) and the least absolute shrinkage and selection operator (LASSO) were used to select the radiomic features. At first, the mRMR feature selection algorithm was performed to rank the radiomic features, and the top 20 features with high relevance were retained. The LASSO logistic regression algorithm was then applied to select the optimized subset of features to construct the radiomic signature. A radiomics score (Rad-score) was calculated for each patient via a linear combination of selected features weighted by their respective coefficients. Calibration curves were plotted to assess the calibration of radiomic signature, accompanied by the Hosmer-Lemeshow test. The clinical model was generated on the basis of clinical factors and CT manifestations using multivariable logistic regression. The area under the curve (AUC) of operating characteristics (ROC), sensitivity, specificity, accuracy, positive predictive value (PPV), and negative predictive value (NPV) were used to assess the predictive performance of radiomic signature and clinical model. Machine learning and artificial intelligence models are prone to overfitting, meaning that they perform well only for images on which they were trained [25]. Therefore, the results of the external test were set for the final assessment.

Results

Clinical and imaging characteristics

A total of 378 patients were included in this study. One hundred and fifty patients had aortic conditions, including the following: 73 thoracic ADs (DeBakey type I, *n* = 15; DeBakey type II, *n* = 9; DeBakey type III, *n* = 49), 25 aortic intramural hematomas, 25 aortic ulcers, 37 aortic aneurysms, 9 aortic thrombi, 5 aortic stenoses, 1 dextroposition of the aortic arch, and 1 right descending aorta were found. There were only 9 DeBakey type II ADs, 8 in the training set and 1 in

Table 1 Clinical and imaging characteristics of patients in the training, validation, and external test sets

Characteristic	Training set			Validation set			External test set		
	TAD	Non-TAD	<i>p</i> value	TAD	Non-TAD	<i>p</i> value	TAD	Non-TAD	<i>p</i> value
Age (mean ± SD)	61.3 ± 17.3	62.1 ± 15.1	0.76	64.2 ± 16.1	60.8 ± 12.6	0.35	49.6 ± 14.3	58.9 ± 17.1	0.07
Gender									
Male	35 (83.3)	115 (66.9)		10 (58.8)	54 (74.0)		12 (85.7)	38 (63.3)	
Female	7 (16.7)	57 (33.1)	0.06	7 (41.2)	19 (26.0)	0.35	2 (14.3)	22 (36.7)	0.11
Chest and/or back pain									
Yes	23 (54.8)	68 (39.5)		13 (76.5)	28 (38.4)		11 (78.6)	25 (41.7)	
No	19 (45.2)	104 (60.5)	0.11	4 (23.5)	45 (61.6)	< 0.05	3 (21.4)	35 (58.3)	< 0.05
Smoking									
Yes	11 (26.2)	63 (36.6)		2 (11.8)	38 (52.1)		6 (42.9)	22 (36.7)	
No	31 (73.8)	109 (63.4)	0.27	15 (88.2)	35 (47.9)	< 0.05	8 (57.1)	38 (63.3)	0.67
Hypertension									
Yes	29 (69.0)	91 (52.9)		12 (70.6)	42 (57.5)		9 (64.3)	32 (53.3)	
No	13 (31.0)	81 (47.1)	0.09	5 (29.4)	31 (42.5)	0.47	5 (35.7)	28 (46.7)	0.46
Displaced intimal flap									
Yes	21 (50.0)	6 (3.5)		8 (47.1)	1 (1.4)		5 (35.7)	1 (1.7)	
No	21 (50.0)	166 (96.5)	< 0.001	9 (52.9)	72 (98.6)	< 0.001	9 (64.3)	59 (98.3)	< 0.001
Displaced calcification									
Yes	14 (33.3)	7 (4.1)		10 (58.8)	4 (5.5)		3 (21.4)	3 (5.0)	
No	28 (66.7)	165 (95.9)	< 0.001	7 (41.2)	69 (94.5)	< 0.001	11 (78.6)	57 (95.0)	< 0.05
High attenuation area									
Yes	10 (23.8)	14 (8.1)		4 (23.5)	0 (0.0)		2 (14.3)	2 (3.3)	
No	32 (76.2)	158 (91.9)	< 0.001	13 (76.5)	73 (100.0)	< 0.001	12 (85.7)	58 (96.7)	0.10

Data are number of patients; data in parentheses are percentage unless otherwise indicated. *Non-TAD* non-thoracic aortic dissection, *SD* standard deviation, *TAD* thoracic aortic dissection

the external test set. The remaining 228 cases were negative on contrast-enhanced CT or MRI. Clinical factors and CT manifestations of patients in the training, validation, and

external test sets are listed in Table 1. The differences of the displaced intimal flaps, displaced intimal calcifications, and high-attenuation areas were significant between thoracic

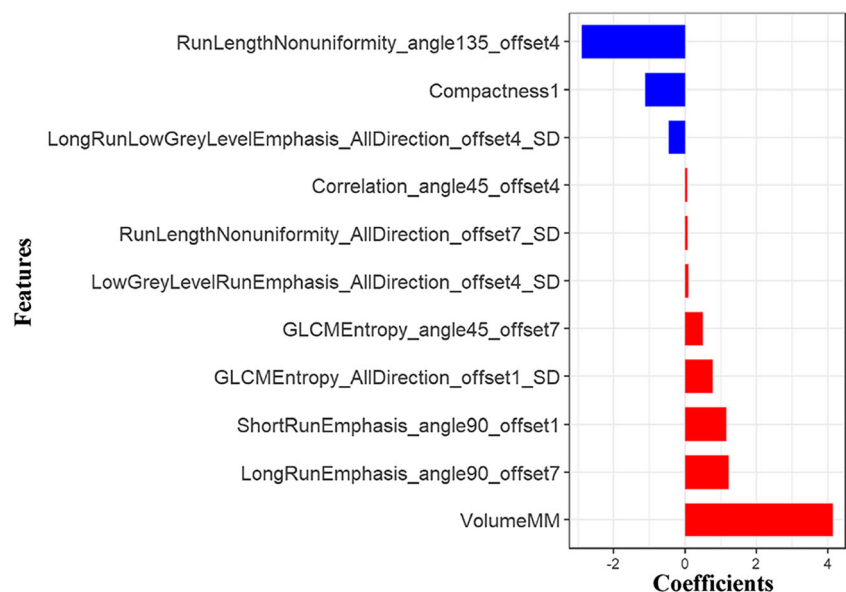
Fig. 3 The most predictive subset of features and the corresponding coefficients

Table 2 The predictive performance of radiomic signature and clinical model in the training, validation, and test sets

	AUC	95% CI	Sensitivity (%)	Specificity (%)	PPV (%)	NPV (%)	Accuracy (%)
Radiomic signature							
Training set	0.91	0.86 – 0.95	92.9	75.6	48.1	97.7	79.0
Validation set	0.92	0.86 – 0.98	94.1	75.3	47.1	98.2	78.9
External test set	0.90	0.82 – 0.98	85.7	91.7	70.6	96.5	90.5
Clinical model							
Training set	0.79	0.72 – 0.87	57.1	95.3	75.0	90.1	87.9
Validation set	0.77	0.65 – 0.90	52.9	87.7	50.0	88.9	81.1
External test set	0.77	0.66 – 0.87	35.7	98.3	83.3	86.8	86.5

AUC area under the curve, CI confidence interval, NPV negative predictive value, PPV positive prediction value

ADs and non-thoracic ADs in the training set (all $p < 0.001$). There were significant differences in the presence of chest and/or back pain, smoking, displaced intimal flaps, displaced intimal calcifications, and high-attenuation areas between thoracic ADs and non-thoracic ADs in the validation set (all $p < 0.05$). The presence of chest and/or back pain, displaced intimal flaps, and displaced intimal calcifications was significantly different between thoracic ADs and non-thoracic ADs in test set as well (all $p < 0.05$).

Radiomic feature selection

Among all the 396 radiomic features, 231 features were found to have reliable inter- and intra-reader agreement for radiomic feature extraction (ICC > 0.75). Among 231 stable radiomic features, 20 features were retained by mRMR. Eleven features with nonzero coefficients were selected using the LASSO logistic regression algorithm on the basis of the training set. The selected features included 2 form factor features, 6 RLM features, and 3 GLCM features. Detailed feature names and corresponding weights are displayed in Fig. 3.

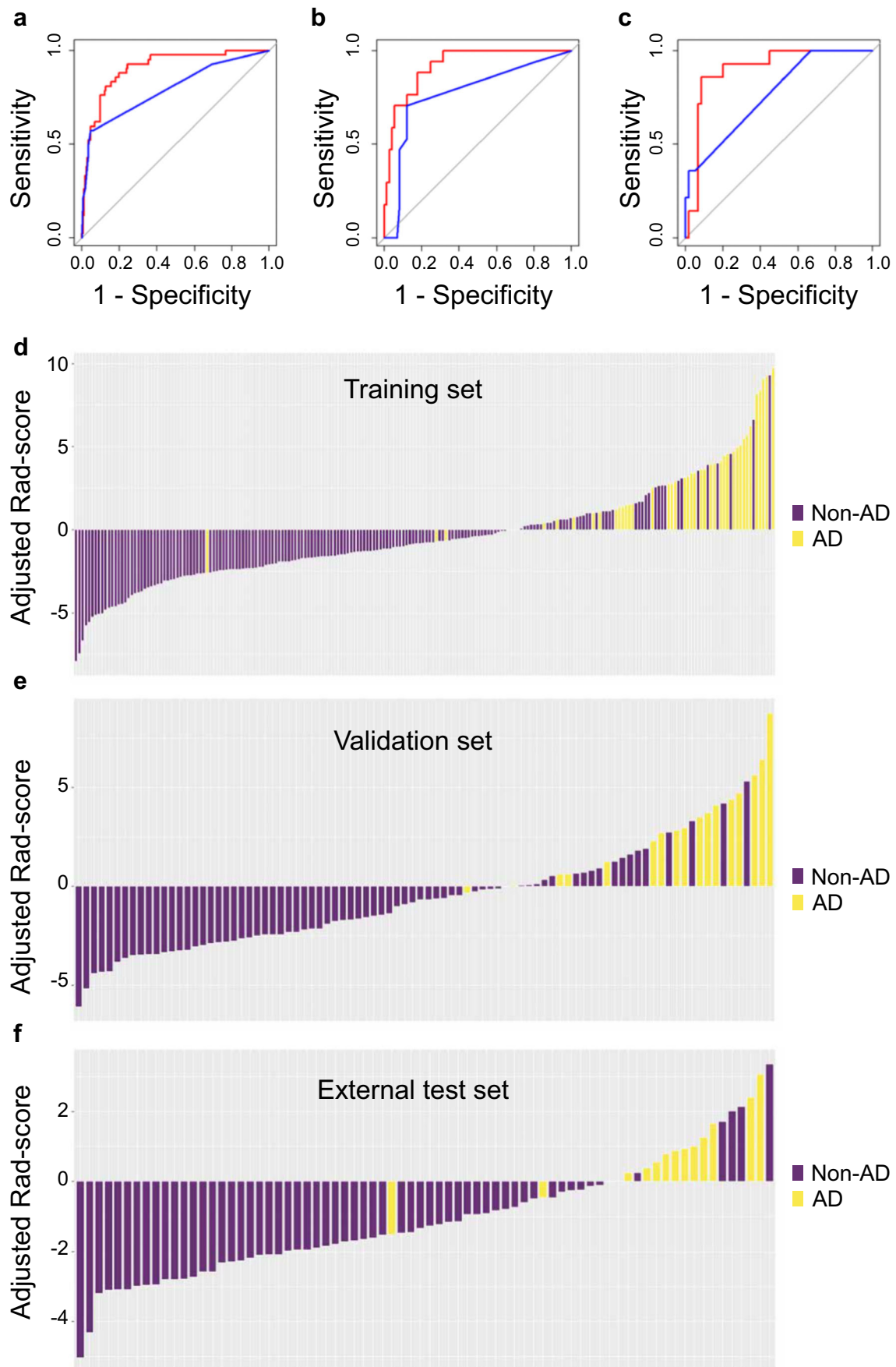
Radiomic signature construction and evaluation

The radiomic signature was constructed via a linear combination of the selected 11 features weighted by their respective coefficients. The formula for the Rad-score is presented in Appendix E2 (online). The cutoff value of radiomic signature was -0.96 . The radiomic signature showed good predictive ability for thoracic ADs in the training, validation, and external test sets. In the training set, the AUC of the radiomic signature was 0.91 (95% confidence interval [CI] 0.86–0.95); the diagnostic accuracy, sensitivity, specificity, PPV, and NPV of the radiomic signature was 79.0%, 92.9%, 75.6%, 48.1%, and 97.7%, respectively. In the validation set, the AUC of the radiomic signature was 0.92 (95% CI 0.86–0.98); the diagnostic accuracy, sensitivity, specificity,

PPV, and NPV of the radiomic signature was 78.9%, 94.1%, 75.3%, 47.1%, and 98.2%, respectively. In the external test set, the AUC of the radiomic signature was 0.90 (95% CI 0.82–0.98); the diagnostic accuracy, sensitivity, specificity, PPV, and NPV of the radiomic signature was 90.5%, 85.7%, 91.7%, 70.6%, and 96.5%, respectively (Table 2; Fig. 4a–c). Although the number of DeBakey type II ADs was small, the predicted diagnosis of these cases was completely consistent with the actual diagnosis. The distributions of the Rad-score for each patient in the three sets are shown in Fig. 4d–f. The calibration curve of the radiomic signature showed good agreement between the predicted and actual diagnosis in the three sets (Fig. 5). In addition, the Hosmer–Lemeshow test also showed good agreement between the predicted and actual diagnosis with p values of 0.26 for the training set, 0.77 for the validation set, and 0.48 for the external test set.

The clinical model was constructed by considering the gender, displaced intimal flaps, and displaced intimal calcifications, with AUCs of 0.79 (95% CI, 0.72–0.87) in the training set, 0.77 (95% CI, 0.65–0.90) in the validation set, and 0.77 (95% CI, 0.66–0.87) in the external test set (Fig. 4a–c). The diagnostic accuracy of the radiomic signature was 87.9%, with a sensitivity of 57.1%, a specificity of 95.3%, a PPV of 75.0%, and a NPV of 90.1% in the training set. The diagnostic accuracy of the radiomic signature was 81.1%, with a sensitivity of 52.9%, a specificity of 87.7%, a PPV of 50.0%, and a NPV of 88.9% in the validation set. The

Fig. 4 ROC curves and waterfall plots of the radiomic signature. **a–c** Comparison of ROC curves between the radiomic signature (red) and clinical model (blue) for the prediction of thoracic AD in the (a) training, (b) validation, and (c) external test sets. **d–f** The Waterfall plot for distribution of Rad-score and aortic diagnosis of individual patients in the (d) training, (e) validation, and (f) external test sets. Rad-score minus cutoff value is the adjusted Rad-score. *Non-TAD* non-thoracic aortic dissection, *Rad-score* radiomics score, *TAD* thoracic aortic dissection



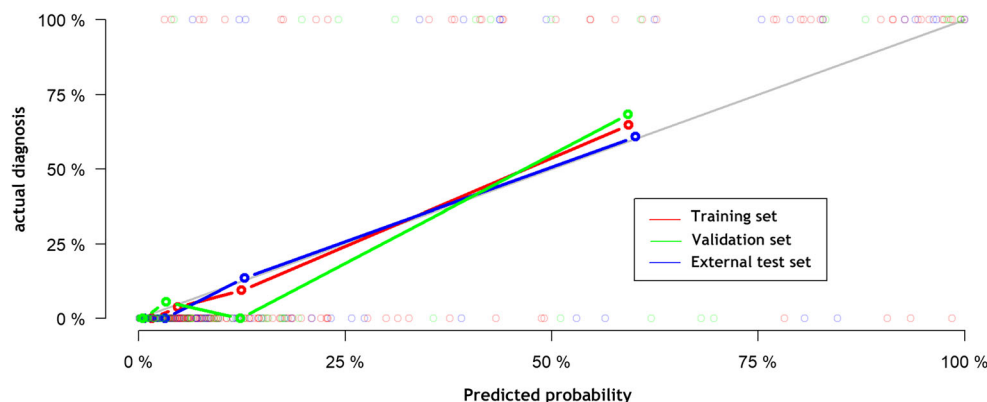


Fig. 5 Calibration curves of the radiomic signature in each set. The calibration curve illustrates the calibration of the radiomic signature in terms of the agreement between predicted and actual diagnosis. The 45° solid gray line represents a perfect prediction, and the other solid line

represents the predictive performance of the radiomic signature in the training (red), validation (green) and external test (blue) sets. The closer a line is to the gray 45° line, the better the predictive accuracy of the radiomic signature

diagnostic accuracy of the radiomic signature was 86.5%, with a sensitivity of 35.7%, a specificity of 98.3%, a PPV of 83.3%, and a NPV of 86.8% in the external test set (Table 2).

The AUCs of clinical model were lower than those of radiomic signature in the training, validation, and external test sets. There were significant differences in the AUCs between the radiomic signature and clinical model in three sets (DeLong test, all $p < 0.05$).

Discussion

In the present study, we developed and validated a radiomic signature that can effectively predict the diagnosis of a thoracic AD on non-contrast CT images. Our radiomic signature showed excellent sensitivity and NPV in the external test set. AD is a serious but treatable disease, and a high sensitivity has profound clinical implications. As shown by the results, the radiomic signature showed better predictive performance compared to the clinical model. In addition, the diagnostic accuracy, specificity, and PPV of the radiomic signature were improved in the external test set.

Despite improvements in diagnostic measures, including imaging techniques and biomarkers, there is still a lack of an efficient, fast, and accurate screening method for ADs. Chest X-ray and transthoracic echocardiography (TTE) can be chosen for patients with a low suspicion of AD [9]. However, rapid chest X-ray has marginal sensitivity and specificity for detecting AD. The poor visualization of the aortic arch and the descending aorta are obvious limitations of TTE [26], not to mention the operator-dependent image quality. A systematic review showed that the diagnostic accuracy of transesophageal echocardiography (TOE), contrast-enhanced CT, and MRI for thoracic AD was sufficient, but leaves advancements to be desired [8]. The accuracy of TOE depends on the experience of the observers, and observer bias can be easily generated when data are calculated

and interpreted by an inexperienced observer [27]. Although MRI is considered the most accurate diagnostic technique for AD, it is rarely recommended as an initial test due to time delay, incompatibility with implanted metal devices, or difficulty in monitoring patients during examination [1]. According to the study of the ESC, contrast-enhanced CT is recommended for the diagnostic imaging modalities of AD [9]. However, one must weigh the pros and cons of ionizing radiation with the probably of AD before suggesting imaging. Some CT manifestations may indicate AD on non-contrast images, such as a displaced intimal flap, displaced intimal calcification, and high-attenuation area [28–30], but these non-contrast CT manifestations are non-specific to AD [31]. Furthermore, the subjective bias and aortic motion artifacts also affect the diagnosis of thoracic AD. Therefore, the use of radiomics to improve the screening ability of non-contrast CT in thoracic AD is valuable.

In this study, the radiomic signature showed a far greater predictive performance compared to the clinical model, possibly due to the ability of radiomic features to model the heterogeneity of blood appearance on non-contrast CT images. Thus, the blood heterogeneity in different flow patterns may be quantified by radiomics on non-contrast CT images. Multiple elements in the whole blood act to contribute to the heterogeneity of blood density [21]. The physiological parameters and processes of blood flow are drastically affected by this heterogeneity and act to promote turbulent flow [19, 20]. In the normal aorta, the flow pattern of the proximal arch and the ascending aorta can maintain a streamlined flow from the distal portion of the arch to the descending thoracic aorta [32]. The pathophysiology of an aortic dissection follows ensuing a tear in the aortic intimal wall, which results in altered blood flow around and in between the exposed layers [33]. The turbulent flow results in further separation of the aortic wall layers and the intimal flap, the latter being the interface between the true and false lumen [20]. There was a significant difference in blood heterogeneity between thoracic ADs and non-thoracic ADs, which may be due to differences in blood flow.

There were only 9 DeBakey type II ADs in this study, but the predicted diagnosis of these cases was consistent with the actual diagnosis. It indicated that placing ROI at the carina plane of descending aorta may be more conducive to radiomic analysis. Although DeBakey type II ADs affect only the ascending aorta, its influence on the flow pattern of ascending aorta may continue to significantly affect the flow pattern of the aortic arch and the descending aorta due to the direction of the blood flow. Compared with the normal aortic arch and ascending aorta, the flow pattern of the normal descending aorta is more stable [32, 34]. If the ROI is placed in the ascending aorta, the flow pattern of this ROI may be less affected by DeBakey type III ADs due to the direction of blood flow. However, the radiomic analysis of DeBakey type II AD requires further research with a larger sample size.

AD is often misdiagnosed without a high clinical index of suspicion [1, 10]. The abrupt onset of severe chest and/or back pain was a typical symptom for thoracic AD. Acute aortic chest pain can mimic other acute emergencies, such as myocardial infarction or pulmonary embolism [10]. However, only 0.09% of chest pain complaints were found to be caused by a thoracic AD [35]. Although approximately one-third of patients presented with painless thoracic ADs in this study, the predictive performance of our radiomic signature was satisfactory. Furthermore, our radiomic signature can effectively screen for thoracic ADs in the absence of clinical information. Therefore, the radiomic signature proposed herein can be applied to all patients undergoing a thoracic CT scan. For patients identified as at risk for thoracic AD by our algorithm, supplemental contrast-enhanced CT or MRI should be used to confirm the diagnosis and prevent unnecessary treatment stemming from false positives.

This study has several limitations. Firstly, there may be a potential selection bias in this study. All patients collected in this study underwent non-contrast chest CT, contrast-enhanced chest CT, or MRI examinations, which may lead to a high positive rate of thoracic AD. Furthermore, there were only 9 DeBakey type II thoracic ADs in this study. Therefore, further studies are needed to increase the sample size and verify the results. While the performance of our two-dimensional feature-based radiomic signature was satisfactory, a three-dimensional feature-based radiomic signature may improve the prediction ability of the radiomic signature. In our view, even if the thoracic AD did not involve aorta at the carina plane, the blood heterogeneity at this plane may be affected by thoracic ADs. Three-dimensional ROI needs careful consideration because of the motion artifacts and complex aortic flow patterns. Finally, the CT images collected in this study were from six different machines. While it is likely that the use of different imaging systems may affect the performance of the radiomic signature, such a reality cannot be avoided in clinical practice.

In conclusion, our study developed a non-contrast CT-based radiomic signature that showed favorable ability in predicting thoracic ADs. Our radiomic signature can be applied to all patients undergoing thoracic CT scan without the burden of additional imaging examinations and subjective CT image interpretations. Therefore, this radiomic signature may be a potential tool for the extensive screening of thoracic ADs and would be of great benefit to patient outcomes.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00330-021-07768-2>.

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Compliance with ethical standards

Guarantor The scientific guarantor of this publication is Zhichao Sun.

Conflict of interest The authors of this manuscript declare relationships with the following companies: GE Healthcare. Peipei Pang contributed to the development of radiomics models described in the study.

Statistics and biometry One of the authors (Peipei Pang) has significant statistical expertise.

Informed consent Written informed consent was waived by the Institutional Review Board.

Ethical approval Institutional Review Board approval was obtained.

Methodology

- Retrospective
- Diagnostic or prognostic study
- Multicenter study

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